

ERIC IRWIN

AI Software Engineer Lead | LLM/RAG Systems | Internal AI Platforms | Technical Leadership

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PROFESSIONAL SUMMARY

AI-focused software engineering lead with deep experience building internal tools, analytics platforms, RAG-style evidence systems, local LLM workflows, high-performance data pipelines, and reusable software frameworks. Led multiple scrum-sized Boeing development teams and owned architecture, design, code review, delivery, demos, hiring input, mentoring, and cross-functional execution. Strong fit for hands-on AI lead roles that need production-minded software engineering, practical LLM integration, data grounding, validation, and platform thinking.

AI / PLATFORM CAPABILITIES

- LLM / RAG systems: local LLMs, Ollama workflows, embeddings, retrieval-style evidence grounding, prompt templates, structured JSON outputs, validation/retry loops, and self-correction patterns.
- Internal AI tools: built Job Application Factory, a local-first AI application system with evidence library, job discovery, scoring, resume/cover/message generation, validation, tracker, and learning loop.
- AI code generation: built a RAG/self-learning strategy generation system trained on NinjaTrader documentation with natural-language code generation and compile-feedback learning.
- Data platforms: built analytics engines processing 10s of GB in minutes and contributed to systems handling 100s of TB of data.
- Production engineering: Python, C#, C++, Java, SQL, Docker, AWS, Streamlit, SQLite, Playwright, Jinja templates, python-docx, pypdf, structured pipelines, CLI tools, and automated reporting.
- Leadership: architecture ownership, code review, standards, reusable frameworks, scrum master responsibilities, mentoring, onboarding, candidate interviews, hiring input, and performance-review input.

SELECTED AI AND AUTOMATION PROJECTS

- Job Application Factory: local-first AI job application automation platform with Streamlit UI, SQLite database, evidence-grounded resume generation, job polling, job scoring, package generation, banned-claim validation, and application learning loop.
- Evidence-Grounded Resume Generator: imports historical resumes and documents, extracts draft evidence, stages bullets for approval, and prevents unsupported claims from entering generated resumes.
- AI Strategy Generation System: generates NinjaTrader strategy code from natural language using domain documentation, retrieval/grounding, compile-error feedback, and valid/invalid output labeling.
- Analyzer Framework: production-grade analytics/reporting framework for multi-run ingestion, canonical modeling, extensible parsing, deterministic analysis, and automated HTML reporting with 100+ configurable sections.
- High-Performance Market Data Engine: processes 10s of GB of tick-level data in under 10 minutes while evaluating thousands of triggers for pattern detection, simulations, and signal discovery.
- License Server / Trading Product Infrastructure: built licensing infrastructure integrated with Wix, email, and payment services to support controlled distribution of automated trading systems.

PROFESSIONAL EXPERIENCE

Trader Dan LLC | Senior Software Engineer | 2025-Present | Remote / Colorado

- Build automated trading, analytics, and workflow systems for NinjaTrader-based strategy development and execution.
- Developed 40+ automated strategies and supporting tools for evaluation, optimization, reporting, and user workflow automation.
- Built high-throughput pipelines capable of running about 100 strategy backtests per hour.
- Create reusable Python/C# tooling for analysis, strategy generation, reporting, and system orchestration.

Pantheon Trading | Technical Lead / Senior Software Engineer | 2023-2025 | Private Trading Group

- Architected and built an automated trading ecosystem spanning strategy bots, analytics infrastructure, market-data pipelines, execution tooling, and evaluation workflows.
- Built and maintained 100+ automated strategies and indicators with tooling for rapid experimentation and repeatable backtesting.
- Improved backtesting and analytics performance by approximately 90% by replacing manual workflows with scalable automation.
- Developed AI-assisted trading tools, market data pipelines, real-time execution support, and performance-analysis systems.
- Enabled members to secure more than \$3M in prop-firm funding using automated systems where none had previously been funded.

ISSAC Corp LLC | Senior Software Engineer | 2015-2023 | Colorado

- Designed and delivered AWS-based AI, analytics, and data-processing systems handling 100s of terabytes of data.
- Built AI-powered prediction systems and data-driven applications for operational decision support.
- Developed logistics optimization, truck routing, inventory tracking, and event scheduling systems used by operations teams.
- Collaborated with stakeholders and engineering teams to translate operational needs into practical software tools.
- Interviewed candidates, influenced hiring decisions, and contributed to team growth and engineering execution.

The Boeing Company | Lead Software Engineer / Software Engineer | 2001-2015 | Seal Beach, CA / Colorado Springs, CO

- Led multiple scrum-sized development teams, typically 3 to 7 members, with responsibility for architecture, design, work assignment, standups/status, code review, schedules, and delivery.
- Served as lead and principal architect/developer for Integrated Visualization Framework, building C++/Qt, Flash, and Java frameworks for Common Operational Picture applications.
- Led a 3-person JEBC2 Joint Effects Based Command and Control team as lead architect and designer.
- Led a 3-person Avionics Dashboard team delivering a Boeing Commercial Aircraft ASIC web portal using Java, HTML5, JavaScript, PostgreSQL/SQL, D3.js, HighCharts, and three.js.
- Created engineering standards, reusable frameworks, coding patterns, and technical processes followed by other engineers.
- Interviewed candidates, influenced hiring decisions, onboarded and mentored junior engineers, and provided input on reviews, promotions, staffing, and performance issues.
- Performed requirements translation, architecture/design documentation, interface specification, build-vs-buy analysis, demo planning, proposal material support, stakeholder coordination, and cross-functional delivery.
- Contributed to a Boeing patent related to multi-level security, software distribution, and software execution; patent details to confirm.

EDUCATION

- M.S. Computer Science, Software Engineering emphasis - Colorado Technical University; GPA listed in older resumes as 3.8/4.0.
- Graduate coursework: algorithms, graph theory, performance measurement, operating systems, networking, security foundations, databases, requirements engineering, systems architecture, software design, and software engineering process.
- B.S. Computer Science - University of California, Riverside; GPA 3.9/4.0; graduated summa cum laude.
- A.S. Mathematics - Grossmont College; GPA 4.0/4.0; President of Alpha Gamma Sigma Honor Society; grew active membership to 100+.